

Practice questions

Q1 [22 Pts]: You may have heard of the university's upcoming plans to renovate campus building: **SCIT**. An **initial survey** was conducted by the university admin a few months ago with the goal of **measuring sentiment of BNU students** towards the renovation of the building.

The results of this initial survey are stored in the DataFrame **responses**, which contains information about each student who responded to the survey. The "Response" column indicates whether a student approved ("Yes") or disapproved ("No") of the renovation plans. The few rows of **responses** are displayed below:

	STUDENT NAME	DEPARTMENT	YEAR	RESPONSE
0	Komal Masood	MDSVAD	5	No
1	Vania Khurshid	SCIT	3	No
2	Husna Usman	RHSA	3	Yes
3	Hafsa Usman	RHSA	3	No
4	Bushra Shahid	SMSLASS	3	Yes
5	Hasnain Ahmad	SMC	4	No

Figure 1: The 6 rows of **responses** DataFrame

(a) [2 Pts] What is the target population of the initial survey? Shade in the bubble corresponding to your answer.

- ☐ A. Students who take classes in SCIT
- ☐ B. BNU students
- ☐ C. University Administration
- ☐ D. Students who responded to the initial survey

(b) [2 Pts] We want to conduct a **new study** with the same **target population** as the initial survey. In this new study, we will randomly sample individuals from **responses** DataFrame and contact them to ask follow-up questions. What is the **sampling frame** of this new study?

- ☐ A. Students who take classes in SCIT
- ☐ B. BNU students
- ☐ C. University Administration
- ☐ D. Students who responded to the initial survey

(c) [2 Pts] Rather than sample from all individuals in **responses**, we decide to sample from a subset of the DataFrame containing only AI and Data Science majors. We email sampled individuals and ask them to fill out a form containing our survey questions. What errors or biases could possibly arise from this sampling scheme? Select all that apply.

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- ☐ A. Selection bias
- ☐ B. Chance error
- ☐ C. Response bias
- ☐ D. Non-response bias

Before conducting our new survey, we want to explore the results stored in responses DataFrame. **The following subparts all involving writing code.** For the remainder of this question, assume that pandas and numpy have been imported as pd and np, respectively.

(d) [2 Pts] Select the option(s) below that will produce a Series **indexed by "Response"** that stores the number of students who gave each response in **descending order**.

```
No      4
Yes      2
Name: Response, dtype: int64
```

The desired output based on the previewed DataFrame rows from the previous page is given above. Select all that apply.

- ☐ A. `responses.groupby("Response").size().sort_values()`
- ☐ B. `responses["Response"].values_counts()`
- ☐ C. `responses.groupby("Response").agg("sum")`
- ☐ D. `responses.groupby("Response")["Department"].count() \`
`.sort_values(ascending = False)`

(e) [2 Pts] Students who are in their 4th or 5th year of university will likely graduate before the building renovation. We suspect that these students may respond differently to other students.

To investigate this theory, select the option(s) below that will extract only rows from responses corresponding to students who have a "Year" **value of 4 or 5**. Assume that the "Year" column contains integer values. Select all that apply.

- ☐ A. `responses[(responses["Year"]==4) | (responses["Year"]==5)]`
- ☐ B. `responses.loc[:, (responses["Year"]>3) | (responses["Year"]<6)]`
- ☐ C. `responses.loc[(responses["Year"]>3) | (responses["Year"]<6), :]`

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□ D. `responses.loc[(responses["Year"]>3) & (responses["Year"]<6), :]`

In Parts (f) and (g), you may write as **much code** as you believe is necessary in each blank, provided it follows the given skeleton code.

(f) [6 Pts] Some departments had more respondents than others. We want to see the results from departments that only had a few students responding to the survey.

Fill in the blanks to construct the DataFrame `results_dep`. The `results_dep` DataFrame contains the same columns as `responses`, but **only** includes rows corresponding to departments that had **fewer than two students** in that department respond to the initial survey.

	STUDENT NAME	DEPARTMENT	YEAR	RESPONSE
0	Komal Masood	MDSVAD	5	No
1	Vania Khurshid	SCIT	3	No
4	Bushra Shahid	SMSLASS	3	Yes
5	Hasnain Ahmad	SMC	4	No

Figure 2: `results_dep` based on the previewed rows of `responses`.

`results_dep = responses.groupby(— $\alpha$ —).— $\beta$ — (lambda sf:— $\gamma$ —)`

(i) Fill in blank α :

(ii) Fill in blank β :

(iii) Fill in blank γ :

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(g) [6 Pts] We think of a new research question: could a student's chocolate consumption impact their opinion on the building renovation.

We obtained a new DataFrame, `chocolate`, that contains the chocolate consumption for each student who responded to the initial survey. The first few rows of `chocolate` are displayed below in Figure 3. You may assume that all names are unique.

We **wish to construct a DataFrame** named `result` that contains all of the original columns of `responses` as well as the chocolate consumption for each student. The form of `result` is shown below in Figure 4.

	First Name	CHOCOLATE CONSUMPTION
0	Komal	1.5
1	Vania	20.5
2	Husna	4.1
3	Hafsa	4.0
4	Bushra	4.2
5	Hasnain	1.3

Figure 3: `chocolate.head(6)`

	Student Name	Department	Year	Response	Chocolate Consumption
0	Komal Masood	MDSVAD	5	No	1.5
1	Vania Khurshid	SCIT	3	No	20.5
2	Husna Usman	RHSA	3	Yes	4.1
3	Hafsa Usman	RHSA	3	No	4.0
4	Bushra Shahid	SMSLASS	3	Yes	4.2
5	Hasnain Ahmad	SMC	4	No	1.3

Figure 4: `result.head(6)`

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Fill in the blanks to produce `result` DataFrame (see Figure 4). You may include as **many** arguments to function calls and as **many** uses of `str` accessors as you believe are necessary.

```
responses["First name"] = responses["Student Name"].str.-A -  
result = responses.-B - (-C -).drop(columns = ["First name"])
```

(i) Fill in blank A:

(ii) Fill in blank B:

(iii) Fill in blank C:

Q2 [8 Pts]: We will be using `elections` dataset from lecture.

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```
[1]: # import packages
import pandas as pd
import numpy as np
elections = pd.read_csv('elections.csv')
elections.head(10)
```

	Year	Candidate	Party	Popular vote	Result	%
0	1824	Andrew Jackson	Democratic-Republican	151271	loss	57.210122
1	1824	John Quincy Adams	Democratic-Republican	113142	win	42.789878
2	1828	Andrew Jackson	Democratic	642806	win	56.203927
3	1828	John Quincy Adams	National Republican	500897	loss	43.796073
4	1832	Andrew Jackson	Democratic	702735	win	54.574789
5	1832	Henry Clay	National Republican	484205	loss	37.603628
6	1832	William Wirt	Anti-Masonic	100715	loss	7.821583
7	1836	Hugh Lawson White	Whig	146109	loss	10.005985
8	1836	Martin Van Buren	Democratic	763291	win	52.272472
9	1836	William Henry Harrison	Whig	550816	loss	37.721543

- (a) [4 Pts]: Write a line of code that returns the elections table sorted in descending order by popular vote.
- (b) [4 Pts]: Would calling `sorted.iloc[[0], :]` give the same result as `sorted.loc[[0], :]` ?

CONGRATULATIONS [1 Pts]

Congratulations! You have completed the Midterm.